

Date \_\_\_\_\_  
Page \_\_\_\_\_

Name : Soham Shinde

Class: TYB Sc (IT) - F2

Roll no: TF2264

## Assignment - 2

Q1) Explain Amazon AWS, Microsoft Azure & Google cloud platform (GCP). Compare their ~~mgf~~ major compute, storage & networking services

⇒ i) ~~A~~ Amazon Web Services (AWS):

Amazon's cloud platform offering compute, storage, networking, databases and more as a pay-as-you-use model

Key services:

- EC2 ⇒ Virtual servers
- S3 ⇒ Object storage
- VPC ⇒ Virtual networking
- RDS ⇒ Managed databases

Example: A college hosts its student portal on EC2 instance instead of ~~by~~ buying a physical server.

ii) Microsoft Azure:

Microsoft's cloud platform for compute, storage, networking, databases and application hosting strong for ~~interact~~ microsoft environments

Q3) Explain Identity and Access management (IAM) & Multi-Factor Authentication (MFA) in cloud security

⇒

\* Identity and Access Management (IAM):

a system that manages digital identities and controls access to cloud resources. Answers who is the user and what can they do?

Components:

- Identity - user/application/service (student, teacher, admin)
- Authentication - verifies identity (password, biometrics, OTP)
- Authorization - decides permitted actions
- Access control - enforces least privilege

Example: College results

| User         | Access               |
|--------------|----------------------|
| Admin        | manage all records   |
| Teacher      | Enter/update marks   |
| Student      | view own result only |
| Unauthorized | No access.           |

\* Multi-Factor-Authentication (MFA)

Requires two or more different factors for login, stronger than password alone.

| Factor             | Meaning                  | Examples              |
|--------------------|--------------------------|-----------------------|
| Something you know | info remembered          | Password, PIN         |
| Something you have | physical object          | Phone, security token |
| Something you are  | <del>ph</del> Biological | Fingerprint, Face     |

Benefits: even if password is stolen, the second factor is still needed

Extra security layer, reduces password only risk, protects sensitive accounts

Q4) Explain cloud security challenges, data privacy, compliance and disaster recovery

⇒ Cloud security challenges:

- i) Data breaches - unauthorized access to protected data
- ii) Unauthorized access - weak passwords or wrong permissions
- iii) Data loss - deletion, failure or attacks (backups help)
- iv) Insecure APIs - poorly secured interfaces become attack paths
- v) Misconfiguration - private storage left public by mistake
- vi) Account Hijacking - stolen credentials take over accounts (MFA needed)
- vii) Insider threats - legitimate users misuse access

Data Privacy: protecting personal / sensitive data so it is collected, stored and shared only appropriately.

Examples of sensitive data: personal details, student records, financial/health info, credentials.

Practices: collect only what is needed, restrict access, encrypt data, use strong ~~auth~~ authentication, delete when no longer required.

Following laws, regulations, standards and organizational policies on data protection, storage, access control, monitoring and retention.

Example: a college must obey data-protection rules and institutional policies for student information shared in the cloud

Disaster Recovery (DR) :- restoring systems, applications and data after major failure (hardware crash, cyber attack, natural disaster)

Basic process: - i) Detect problem → Activate recovery

↓  
Resume operations ← Test ← Restore data/system

Backup - a separate copy of data kept so original can be restored. often both cloud and secondary backups are maintained.